

Shyamendra Singh

Sarendhi, Agra, U.P. (India) | +91-7302886011 | shyamendra.me@gmail.com | [LinkedIn](#) | [Portfolio](#)

EDUCATION

B.Tech (Agricultural Engineering)	Dayalbagh Educational Institute, Agra	8.02 CGPA (Till 5 th Sem)	2026 (Pursuing)
Intermediate (XII)	SMT Maharaj Kunwari Inter College, Agra	80.2%	2021
Highschool (X)	SMT Maharaj Kunwari Inter College, Agra	86.5%	2019

PROFESSIONAL EXPERIENCE

Virtual Labs, MoE-NMEICT	July 2023 - Present
• <i>Project Intern</i>	<i>Agra, UP</i>
◦ Co-operating as an intern in Virtual Labs, a Ministry of Education project under NMEICT, at my college.	
◦ Responsible for designing simulations, illustrations, and preparing content of the virtual experiments.	
ICAR - Indian Agricultural Research Institute	June 2024
• <i>Internship Trainee</i>	<i>Pusa, New Delhi</i>
◦ Gained expertise in mechatronics, CAD and 3D printing, applying these skills to prototype tools and equipments for agricultural applications. Developed <i>Wireless BMI Measuring Device</i> .	
Hankernest Technologies Pvt. Ltd.	June 2022
• <i>Internship Trainee</i>	<i>Dharamkot, HP</i>
◦ A training-cum-internship program covering python programming, practical AI/ML (transfer learning system), practical electronics, internet of things, 3D printing, and CAD designing.	

TECHNICAL SKILLS & INTERESTS

• CAD & 3D Modelling: AutoCAD, SolidWorks	• Programming & Scripting: Python, HTML/CSS
• GIS & Spatial Analysis: QGIS, ArcGIS	• Other Tools: Microsoft Office
• Interests: 3D Designing, GIS, 3D Printing, IoT, Machine Learning	

PROJECTS

Wireless BMI Measuring Device (3D CAD & Printing, ESP32, Arduino IDE)
A portable device for wireless measurement of anthropometric dimensions and body mass index using sensors and microcontrollers, advancing ergonomic studies in agricultural settings.
Low-cost Automatic Hydroponic Unit (3D CAD, Manufacturing)
A vertical hydroponic system designed to reduce costs by 70% and optimize space usage by 20%, featuring automation for nutrient management and watering schedules.
Crop Recommendation Web App  (Python, KNN, Flask)
A web application utilizing the k-nearest neighbors (KNN) algorithm for crop recommendation based on soil and climatic parameters, aimed at enhancing decision-making for farmers.
Soil Texture Calculator  (Python, Matplotlib, Flask)
A web-based tool for soil classification according to USDA standards, providing automated textural analysis and generating ternary graphs for visual representation.

CERTIFICATIONS

• Supervised Machine Learning: Regression and Classification course from Stanford University
• Artificial Intelligence for Everyone (AI4E) course from Dayalbagh Educational Institute, Agra
• Rajasthan State Certificate in Information Technology (RSCIT) from VMOU, Kota

HACKATHONS & WORKSHOPS

• Qualified the institution level stage of Smart India Hackathon (SIH 2023, SIH2024)
• Participated in one day online workshop on Remote Sensing based Data Analytics in Agriculture organised by IIRS
• Participated in one-week hands-on workshop on Digital Manufacturing organised by Dayalbagh Educational Institute, Agra